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328453 (28)

BE (4th Semester)

Examination, April-May, 2014

Branch : Et & T

ANALOG ELECTRONICS (NEW)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) All questions are compulsory and carries equal marks.

(ii) Part (a) of each question is compulsory, attempt any two out of (b), (c) and (d).

(iii) Assume suitable data if necessary.

Unit-I

Q. 1. (a) State Miller theorem. 2

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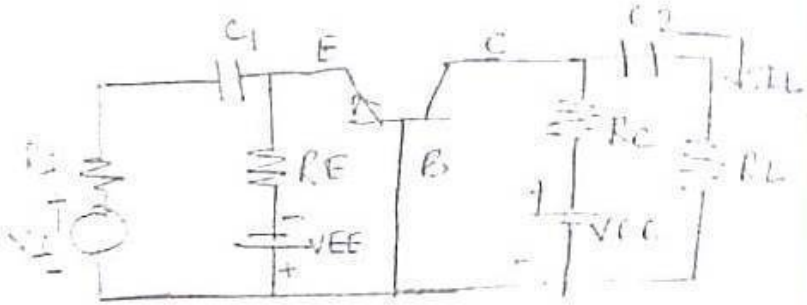
(b) A CB amplifier as shown in fig has following components $R_S = 600\ \Omega$, $R_C = 5.6$ K, $R_1 = 5.6$ K, $R_L = 39$ K. The transistor parameters are hie = 1K, hfe = 85, hoe = 2 μA/V. Calculate Ri, Ro and Av. 7



(c) Explain the operation of FET as VVR 7

(d) Using both exact and approximate analysis derive the expression for (i) Ai (ii) Ri (iii) Av and (iv) Ro for CE amplifier with an emitter resistance 7

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Unit-II

(a) Why h parameter model is not suitable for high frequency signals. 2

(b) Derive an expression for current gain with resistive load for a CE transistor at high frequency. 7

(c) For an emitter follower of the given figure at high frequency, obtain $K = \frac{V_e}{V_i'}$ and given $g = g_m + g_{b'e}$. Also prove that : 7

(i) $K = \frac{gR_L}{1+gR_L}\left[\frac{1+\frac{j\omega C_{ce}}{g}}{1+j\omega\frac{(C_{ce}+C_L)R_L}{1+gR_L}}\right]$

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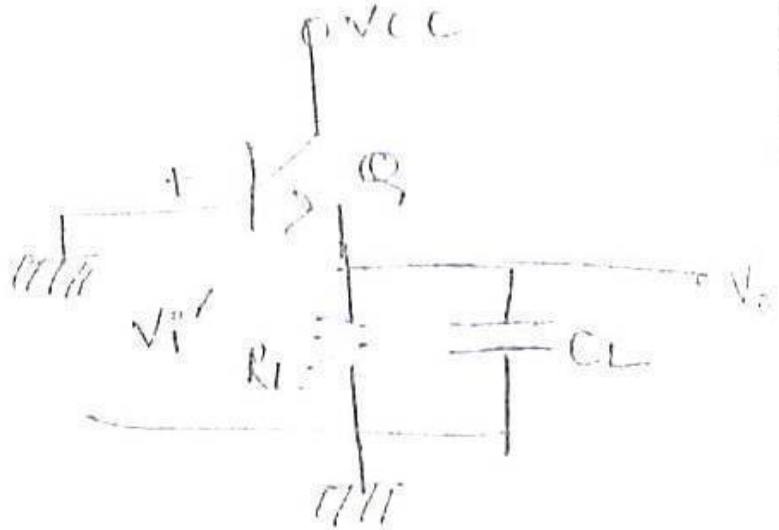
(ii) [illegible]



(d) The following low frequency h parameters are known for given transistor amplifier: $I_C = 10$ mA and $V_C = 10$ V at room temperature. $h_{ie} = 500\,\Omega$, $h_{re} = 10^{-4}$, $h_{oe} = 4 \times 10^{-5}$ A/V

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the same operating point I_C = 50 μA, C_μ = 3 pF. Calculate the values of all hybrid-π parameters. Assume V_T = 26 mV and hfe = 100. 7

Unit-III

Q. 3. (a) What are the different types of distortion that may exist in amplifier. 2

(b) Explain in detail the effect of cascading on B.W. and Rise time. 7

(c) The bandwidth of an amplifier extends from 20 Hz to 20 kHz. Find the frequency range over which voltage gain down less than 1 dB from its mid band value. 7

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(d) Differentiate between RC coupling, transformer coupling and direct coupling. 7

Unit-IV

Q. 4. (a) What are the basis on which feedback amplifiers are classified. 2

(b) Derive input and output resistance for current series feeback. 7

(c) An amplifier consists of three identical stages connected in cascade. The output voltage is sampled and returned to the input in series opposing. If it is specified that relative change $\Delta A_f / A_f$ in closed-loop voltage gain A_f must not exceed γ_f

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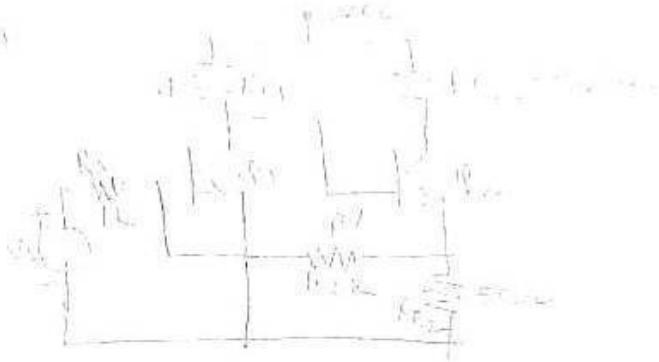
[illegible] the [illegible] of the open loop gain A of amplifier is [illegible]

$$A = [illegible]$$

where [illegible] change in Voltage gain of open loop and the amplifier ?

(d)



For the circuit shown identify the topology and assume hie = 1.1 K, hfe = 50 then calculate [illegible] ?

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Unit-V

Q. 5 (a) What is Barkhausen criteria. 2

(b) Explain RC phase shift oscillator and also derive the expression for operating frequency. 7

(c) Explain the working principle of crystal oscillator with the help of neat circuit diagram. State the frequency stability criterion for crystal oscillator. 7

(d) Write short note on Hartley oscillator. 7

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